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18/166,653	02/09/2023	Masahiro TAKAHASHI	223566	9371

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EXAMINER	
ALMADHRHI, WESAM NMN	

ART UNIT	PAPER NUMBER
3666	

NOTIFICATION DATE	DELIVERY MODE
09/12/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/166,653	Applicant(s) TAKAHASHI, Masahiro	
	Examiner WESAM N ALMADHRHI	Art Unit 3666	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 1,3-7,11-19.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,3-7 and 11-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,3-7 and 11-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Arguments/Amendments

The amendment filed June 16th, 2023 has been entered. Claims 1-20 are currently pending in the Application.

Applicant's arguments with respect to the rejection of claims under 35 U.S.C 103 have been considered but are moot because the new ground of rejection does not rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1, 11, 12, 15, and 17 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent Publication No. 20200294394, to Guoet el al. (hereinafter Guo), and further in view of U.S. Patent Publication No. 20210056840, to Yamazaki et al (hereinafter Yamazaki).

Regarding claim 1, Guo teaches **A control system comprising a processor configured to: control a plurality of autonomous mobile objects** (See at least paragraph [Abstract] "system groups vehicles traveling"). Further, (See at least paragraph [0084] "the vehicle 1001 can be an autonomous or semi-autonomous vehicle"). **that are configured to autonomously move in a group from a departure point to a destination point along a route;** (See at least paragraph [0021] "vehicle grouping is based on the estimated time of vehicle arriving at the control zone").

allocate each of the autonomous mobile objects to either a first group or a second group; (See at least paragraph [0059] "first group (cluster) entering the control zone is early than the time the first vehicle of the second group (cluster)").

Guo fails to explicitly disclose, However, Yamazaki discloses, **identify a dividing element on the route;** (See at least paragraph [Abstract] "A traffic signal control apparatus capable of controlling signal light colors at a target intersection").

depart the first group from the departure point; (See at least paragraph [0023] "the platoon vehicles to preferentially pass through the target intersection by extending a clearance interval of the target intersection.").

determine whether a predetermined time period has elapsed after the first group has departed from the departure point, the predetermined time period being a sum of a first time period and a second time period; and in a case where the predetermined time period has elapsed after the first group has departed from the departure point, depart the second group from the departure point, (See at least paragraph [0109] "When the determination result in step ST**12** is negative (when inequality (2) is not satisfied), the control unit **401** generates "second message" that notifies the leading vehicle **5**A

to stop at the stop line, and transmits the second message to the roadside communication apparatus **3** (step ST**15**). Therefore, the second message is downlink-transmitted to the roadside communication apparatus **3**. The control unit **401** ends the processing after execution of step ST**15**."). Further, (See at least paragraph [0112] "When the determination result in step ST**12** is positive (when inequality (2) is satisfied), the control unit **401** extends the yellow interval Y by a predetermined time AY so that the tail-end vehicle **5**D can pass the stop line of the intersection J by the time the yellow light finishes (step ST**13**), and thereafter executes the process of "notifying that passing is possible" in step ST**14**.").Furthermore, (See at least paragraph [0101] "When the platoon vehicles **5**P are traveling on the inflow road of the intersection J**1**, the control unit **401** determines whether or not the tail-end vehicle **5**D of the platoon vehicles **5**P can pass a stop line of the intersection J by the time the remaining green interval Gr of the intersection J**1** elapses (step ST**11**).").

wherein the dividing element is in either a first condition or a second condition, (See at least paragraph [0010] "a traffic signal control method for controlling signal light colors at a target intersection"). **the first condition continuing for the first time period and the second condition continuing for the second time period,** (See at least paragraph [0109] "When the determination result in step ST12 is negative (when inequality (2) is not satisfied), the control unit 401 generates "second message" that notifies the leading vehicle 5A to stop at the stop line, and transmits the second message to the roadside communication apparatus 3 (step ST15). Therefore, the second message is downlink-transmitted to the roadside communication apparatus 3. The control unit 401 ends the processing after execution of step ST15."). Further, (See at least paragraph [0112] "When the determination result in step ST12 is positive (when inequality (2) is satisfied), the control unit 401 extends the yellow interval Y by a predetermined time AY so that the tail-end vehicle 5D can pass the stop line of the intersection J by the time the yellow light finishes (step ST13), and thereafter executes the process of "notifying that passing is possible" in step ST14.").Furthermore, (See at least paragraph [0101] "When the platoon

vehicles 5P are traveling on the inflow road of the intersection J1, the control unit 401 determines whether or not the tail-end vehicle 5D of the platoon vehicles 5P can pass a stop line of the intersection J by the time the remaining green interval Gr of the intersection J1 elapses (step ST11).”). Further (See at least paragraph [0056-0057] “predetermined cycle (e.g., 1 min), sensor information measured by roadside sensors such as vehicle detectors and image sensors (not shown), and calculates, in each predetermined cycle (e.g., 2.5 min), a traffic index such as link travel time, based on the received sensor information. The central apparatus 4 can perform traffic actuated control in which signal control parameters (split, cycle length, offset, and the like) at each intersection J are adjusted based on the calculated traffic index. ”). **in a case where the dividing element is in the first condition, the autonomous mobile objects pass through the dividing element, and in a case where the dividing element is in the a second condition, the autonomous mobile objects stop at the dividing element.** (See at least paragraph [0045-0046] “ Green means that a vehicle can go straight ahead, turn left, and turn right at an intersection. Yellow means that a vehicle should not advance over a stop position (excluding a case where the vehicle cannot safely stop at the stop position). Red means that a vehicle should not advance over a stop position. Therefore, green is a light color indicating that a vehicle traveling on an inflow road of an intersection has right-of-way at the intersection. Red is a light color indicating that the vehicle traveling on the inflow road of the intersection does not have right-of-way at the intersection. Yellow is a light color indicating that the vehicle does not have right-of-way in principle, but has right-of-way only when the vehicle cannot safely stop at the stop position.”).

Guo as modified by Yamazaki, are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo that discloses platoon vehicle systems to incorporate the teachings of Yamazaki for the purposes of the traffic control light to aid the two groups in order to aid in road congestion.

Regarding claim 11, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo fails to explicitly disclose, However, Yamazaki discloses, **wherein the processor is configured to move the second group directly and uninterruptedly from the departure point to the destination point without stopping the second group on the route that includes the dividing element.** (See at least paragraph [0012] "When the determination result in step ST**12** is positive (when inequality (2) is satisfied), the control unit **401** extends the yellow interval Y by a predetermined time AY so that the tail-end vehicle **5**D can pass the stop line of the intersection J by the time the yellow light finishes (step ST**13**), and thereafter executes the process of "notifying that passing is possible" in step ST**14**.").

Guo as modified by Yamazaki, are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo that discloses platoon vehicle systems to incorporate the teachings of Yamazaki for the purposes of the traffic control light to aid the two groups in order to aid in road congestion.

Regarding claim 12, and commensurate claims 13, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo fails to explicitly disclose, However, Yamazaki discloses, **wherein the dividing element is a traffic light.** (See at least paragraph [0049] "the traffic signal control system").

Guo as modified by Yamazaki, are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo that discloses platoon vehicle systems to incorporate the teachings of Yamazaki for the purposes of the traffic control light to aid the two groups in order to aid in road congestion.

Regarding claim 15, and commensurate claims 17, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo discloses, **wherein the number of the plurality of the autonomous mobile objects is seven, and the processor is configured to allocate four of the plurality of the autonomous mobile objects to the first group and three of the plurality of the autonomous mobile objects to the second group.** (See at least paragraph [0015] " the vehicles are grouped into groups (clusters) and the vehicles of different groups are prevented to travel within the control zone concurrently in the sense that the first vehicle of the following group (cluster) cannot pass the last vehicle of the preceding group (cluster). In such a way, the number of vehicles within the control zone is limited. When the vehicles are in the control zone, the embodiments determine the motion trajectory and motion strategy for these vehicles in the same group. Because the number of vehicles in the control zone are limited, the embodiments ensure the safety and optimality of passing the intersection.").

Claims 3,4, and 5 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent Publication No. 20200294394, to Guoet el al. (hereinafter Guo), and further in view of U.S. Patent Publication No. 20210056840, to Yamazaki et al (hereinafter Yamazaki), and further in view of U.S. Patent Publication No. 20190250639, to Xu et al. (hereinafter Xu).

Regarding claim 3, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo fails to explicitly disclose, However, Xu discloses, **wherein the processor is configured to calculate number of the autonomous mobile objects that are able to pass through the dividing element at once during the first time period, based on a length of the first time period, a distance of a passing section in the dividing element, a moving speed of the autonomous mobile object, and a shape of the group.** (See at least paragraph [0091] "the vehicle platoon crosses the intersection 66 together as a group without any dissociation. In one example, the lead vehicle may wait for confirmation (e.g., receipt of a confirmation message) from the intersection controller 30 or a predetermined number of nearby vehicles before leading the vehicle platoon across the intersection 66 without any disconnection of the

platoon of vehicles. This prevents the platoon of vehicles from being separated by other vehicles entering the intersection. It also improves the speed of traffic flow because multiple stops need not be made by the platoon of vehicles"). Further, (See at least paragraph [0090] "The platoon of vehicles stops at the stop sign 67. The lead vehicle may pause to allow the gap distance between vehicles to close"). Further, (See at least paragraph [0091] "the speed of traffic flow because multiple stops need not be made by the platoon of vehicles"). Further, (See at least paragraph [0030] "The vehicle platoon control system allows the vehicle platoon to maintain a cohesive group or formation by predicting the state of intersections and adjusting the vehicle platoon so that the intersection does not interrupt the group or formation.").

Guo as modified by Xu, are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo as modified by Yamazaki to incorporate the teachings of Xu for the purposes of the traffic control light to aid the two groups in order to aid in road congestion.

Regarding claim 4, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo fails to explicitly disclose, However, Xu discloses, **the dividing element includes a space that is configured to store the autonomous mobile objects, and the processor is configured to calculate a number of the autonomous mobile objects that are able to be stored in the space based on a size of the space, a size of each of the autonomous mobile objects that fits into the space, and a shape of the group.** (See at least paragraph [0083] "an example intersection with a ramp 60 connecting two roadways. The ramp may connect a major artery (e.g., highway, interstate, freeway) to a collector road or a minor road. The ramp 60 may be an off ramp that gradually veers from the major artery to the minor road, which may be perpendicular to the major artery. Congestion on the ramp 60 may be subject

to a traffic signal or a stop sign with the minor road. In addition, congestion on the minor road may easily back up to the major road."). Further**,** (See at least paragraph [0030] "The vehicle platoon control system allows the vehicle platoon to maintain a cohesive group or formation by predicting the state of intersections and adjusting the vehicle platoon so that the intersection does not interrupt the group or formation.").

Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo as modified by Yamazaki to incorporate the teachings of Xu for the same motivation reasons in claim 3.

Regarding claim 5, Guo as modified by Yamazaki and Xu disclose the claimed features of claim 3, Guo fails to explicitly disclose, However, Xu discloses, **wherein processor is configured to allocate the autonomous mobile objects to either the first group or the second group such that a number of the autonomous mobile objects in both the first group and the second group is equal to or less than the number of the autonomous mobile objects that are able to pass through the dividing element at once during the first time period.** (See at least paragraph [0034] [0059] "control an autonomous vehicle platoon system, or another type of platoon, to successfully drive through these road geometries."). Further, (See at least paragraph [0091] "the vehicle platoon crosses the intersection 66 together as a group without any dissociation. In one example, the lead vehicle may wait for confirmation (e.g., receipt of a confirmation message) from the intersection controller 30 or a predetermined number of nearby vehicles before leading the vehicle platoon across the intersection 66 without any disconnection of the platoon of vehicles. This prevents the platoon of vehicles from being separated by other vehicles entering the intersection. It also improves the speed of traffic flow because multiple stops need not be made by the platoon of vehicles").

Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo as modified by Yamazaki to incorporate the teachings of Xu for the same motivation reasons in claim 3.

Claims 6 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent Publication No. 20200294394, to Guoet el al. (hereinafter Guo), and further in view of U.S. Patent Publication No. 20210056840, to Yamazaki et al (hereinafter Yamazaki), and further in view of U.S. Patent Publication No. 20190250639, to Xu et al. (hereinafter Xu), and further in view of U.S. Patent Publication No. 20170270785, to Umehara et al (hereinafter Umehara).

Regarding claim 6, Guo as modified by Yamazaki and Xu disclose the claimed features of claim 4, Guo fails to explicitly disclose, However, Umehara discloses, **wherein the processor is configured to allocate the autonomous mobile objects to either the first group or the second group such that a number of the autonomous mobile objects in both the first group and the second group is equal to or less than the number of the autonomous mobile objects that the dividing element is able to store in the space.** (See at least paragraph [0053] "the roadside control apparatus of the present embodiment, the control unit preferably transmits a convoy delimiting command for separating a mobile body platoon that is composed of a plurality of mobile bodies equipped with the mobile communication apparatus and is traveling in a convoy, into a platoon of preceding vehicles that can pass through the intersection at a green light this time and a platoon of following vehicles that cannot pass through the intersection at the green light this time, to the mobile communication apparatus of the mobile body platoon"). Further, (See at least paragraph [0258] "In the figures, a reference numeral 43Gx denotes "preceding vehicles", in a vehicle platoon 43G traveling in a convoy, to be allowed to pass through an intersection Jk at the present green interval, and a reference numeral 43Gy denotes "following vehicles", in the vehicle platoon 43G, to be allowed to pass through the intersection Jk at the next green interval").

Guo as modified by Umehara, are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo as modified by Yamazaki and Xu to incorporate the teachings of Umehara for the purposes of the traffic control light to aid the two groups in order to aid in road congestion.

Claims 7, 17, and 19 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent Publication No. 20200294394, to Guo et el al. (hereinafter Guo), and further in view of U.S. Patent Publication No. 20210056840, to Yamazaki et al (hereinafter Yamazaki, and further in view of U.S. Patent Publication No. 20200372805, to Kim et al (hereinafter Kim).

Regarding claim 7, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo fails to explicitly disclose, However, Kim discloses, **wherein the processor is configured to allocate the autonomous mobile objects in an order in which a remaining amount of a battery is low.** (See at least paragraph [0014] "a method of controlling autonomous platooning of an apparatus for controlling autonomous platooning of an electric vehicle including an index assignment unit, a driving arrangement order determination unit, and a controller includes, upon receiving an autonomous platooning control request signal, assigning respective indexes to autonomous platooning vehicles, by the index assignment unit, acquiring battery information from the vehicles to which the indexes are assigned, and calculating respective battery state-of-charge (SoC) values of the autonomous platooning vehicles based on the acquired battery information, by the driving arrangement order determination unit, calculating a difference value between a first battery SoC value of a following vehicle having a maximum battery SoC value and a second battery SoC value of a lead vehicle among the autonomous platooning vehicles, by the driving arrangement order determination unit, checking whether the calculated difference value is greater than a threshold value, by the driving arrangement order determination unit, when the

calculated difference value is greater than the threshold value, determining the driving arrangement order to arrange the following vehicle having the maximum battery SoC value at a frontmost position of the autonomous platooning vehicles, by the driving arrangement order determination unit, and controlling the vehicles to rearrange the autonomous platooning vehicles according to the determined driving arrangement order, by the controller.").

Guo as modified by Kim, are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo that discloses platoon vehicle systems to incorporate the teachings of Kim that discloses SOC platoon positioning in order for the platoon formation leader guide the remaining vehicles.

Regarding claim 17, and commensurate claims 19, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo fails to explicitly disclose, However, Kim discloses, **wherein a remaining amount of each battery in the autonomous mobile objects in the first group is smaller than a remaining amount of each battery in the autonomous mobile objects in the second group.** (See at least paragraph [0014] "a method of controlling autonomous platooning of an apparatus for controlling autonomous platooning of an electric vehicle including an index assignment unit, a driving arrangement order determination unit, and a controller includes, upon receiving an autonomous platooning control request signal, assigning respective indexes to autonomous platooning vehicles, by the index assignment unit, acquiring battery information from the vehicles to which the indexes are assigned, and calculating respective battery state-of-charge (SoC) values of the autonomous platooning vehicles based on the acquired battery information, by the driving arrangement order determination unit, calculating a difference value between a first battery SoC value of a following vehicle having a maximum battery SoC value and a second battery SoC value of a lead vehicle among the autonomous platooning vehicles, by

the driving arrangement order determination unit, checking whether the calculated difference value is greater than a threshold value, by the driving arrangement order determination unit, when the calculated difference value is greater than the threshold value, determining the driving arrangement order to arrange the following vehicle having the maximum battery SoC value at a frontmost position of the autonomous platooning vehicles, by the driving arrangement order determination unit, and controlling the vehicles to rearrange the autonomous platooning vehicles according to the determined driving arrangement order, by the controller.").

Guo as modified by Kim, are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo that discloses platoon vehicle systems to incorporate the teachings of Kim that discloses SOC platoon positioning in order for the platoon formation leader guide the remaining vehicles.

Claims 14, and 15 is/are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Patent Publication No. 20200294394, to Guo et el al. (hereinafter Guo), and further in view of U.S. Patent Publication No. 20210056840, to Yamazaki et al (hereinafter Yamazaki, and further in view of U.S. Patent Publication No. US2021188594A1, to HSU et al (hereinafter HSU).

Regarding claim 14, and commensurate claims 15, Guo as modified by Yamazaki disclose the claimed features of claim 1, Guo fails to explicitly disclose, However, HSU discloses, **wherein the dividing element is an elevator.** (See at least paragraph [Abstract] "operating a shuttle elevator group including.").

Guo as modified by HSU , are analogous art because they are in the same field of endeavor, platoon systems. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified the system of Guo that discloses platoon

vehicle systems to incorporate the teachings of HSU for the purposes of controlling vehicles in a building.

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Wesam Almadhrhi whose telephone number is (571) 270-3844. The examiner can normally be reached on 7:30 AM - 5PM Mon-Fri Eastern Alt Fri.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Anne Antonucci can be reached on (313) 446-6519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-

direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/WESAM NMN ALMADHRHI/
Examiner, Art Unit 3666

/ANNE MARIE ANTONUCCI/
Supervisory Patent Examiner, Art Unit 3666